

CORE ANALYSIS REPORT

**ANGSI-1
MALAYSIA**

**R.C. SOMMER, A.C. WOOD
DECEMBER 1976**

ESSO EXPLORATION MALAYSIA INC.

CORE ANALYSIS REPORT: ANGSI
NO.1



CORE ANALYSIS REPORT

ANGSI WELL NO. 1

ESSO EXPLORATION MALAYSIA INC.

ESSO PRODUCTION MALAYSIA INC.

R000006



INFORMATION CENTRE
EXPLORATION DATA

EXXON PRODUCTION RESEARCH COMPANY

CORE ANALYSIS REPORT
ANGSI WELL NO. 1
ESSO EXPLORATION MALAYSIA INC.

R. C. Sommer
A. C. Wood

Production Engineering Division

December 1974

EPR.114PS.74

FOR ESSO EXPLORATION MALAYSIA INC. USE ONLY

This report contains Esso Exploration Malaysia information; copies should not be given to companies other than Exxon Corporation without Esso Exploration Malaysia's written consent.

TABLE OF CONTENTS

Tables

Summary of Porosity and Permeability Data	I
Summary of Centrifuge Gas-Oil Capillary Pressure Data	II
Summary of Electrical Property Measurements	III

Figures

Gas-Oil Capillary Pressure Curves	1 through 4
Resistivity Ratio Vs. Water Saturation	5 through 8
Formation Factor Vs. Porosity	9

TABLE I

Summary of Porosity and Permeability DataAngsi Well No. 1 ϕ No 2

Depth, Ft.	Plug No.	Porosity, %	Gas Permeability, md	Density, GM/CC	
				Bulk	Grain
8012	AN-28	16.3	4.42	2.23	2.66
8013	AN-29	14.1	2.45	2.30	2.68
8015	AN-30	16.2	19.4	2.23	2.66
8016	AN-31	15.4	7.99	2.26	2.67
8029	AN-32	9.5	0.10	2.44	2.69
8032	AN-33	14.2	6.00	2.30	2.69
8034	AN-34	17.0	50.2	2.22	2.67
8036	AN-35	14.9	23.3	2.28	2.68
8037	AN-36	14.4	9.17	2.30	2.68
8038	AN-37	15.9	51.8	2.24	2.66
8039	AN-38	15.3	23.5	2.26	2.67
8040	AN-39	16.4	37.1	2.24	2.68

up to 50

Siltst

Siltst

Siltst

TABLE II
Summary of Centrifuge Gas-Oil
Capillary Pressures Data

<u>Capillary Pressure, PSI</u>	<u>Avg. Liquid Saturation % PV</u>	<u>Capillary Pressure, PSI</u>	<u>Avg. Liquid Saturation % PV</u>
<u>Core No. AN-30, 8015 Ft.</u>		<u>Core No. AN-36, 8037 Ft.</u>	
2.4*	100	3.0*	100
5.7	32.9	5.8	37.2
23.3	25.3	23.5	30.4
51.3	25.3	51.8	29.5
90.8	22.8	91.8	27.1
<u>Core No. AN-37, 8038 Ft.</u>		<u>Core No. AN-39, 8040 Ft.</u>	
1.9*	100	2.1*	100
5.8	22.8	5.8	28.5
23.6	18.5	23.5	23.2
51.9	18.3	51.8	22.5
91.9	17.6	91.7	19.6

*Estimated

Gas-Oil Surface Tension = 28.0 Dynes/CM

TABLE III
Summary of Electrical Property Measurements

		<u>Water Saturation*</u> <u>% PV</u>	<u>Resistivity Ratio</u> <u>R/Ro</u>
Core No.	AN-28	40.25	7.99
Depth, ft.	8012		
Porosity, %	16.3	25.22	17.17
Permeability, md	4.42		
Formation Factor	42.8		
Saturation Exponent	2.11		
Core No.	AN-29	43.33	6.29
Depth, ft.	8013		
Porosity, %	14.1	28.17	12.09
Permeability, md	2.45		
Formation Factor	58.03		
Saturation Exponent	2.01		
Core No.	AN-32	49.49	4.19
Depth, ft.	8029		
Porosity, %	9.5	41.01	5.67
Permeability, md	0.10		
Formation Factor	137.6		
Saturation Exponent	1.94		
Core No.	AN-38	42.54	9.74
Depth, ft.	8039		
Porosity, %	15.3	26.16	20.95
Permeability, md	23.5		
Formation Factor	43.6		
Saturation Exponent	2.45		

*Laboratory Synthetic Brine (80,000 ppm NaCl + 20,000 ppm KCl)

Figure 1

CAPILLARY PRESSURE CURVE

FIELD Anqsi
RESERVOIR _____
WELL No. 1
DEPTH, ft 8015

CORE AN-30
PERMEABILITY, md 19.4
POROSITY, % 16.2
METHOD Centrifuge: Air-Kerosene

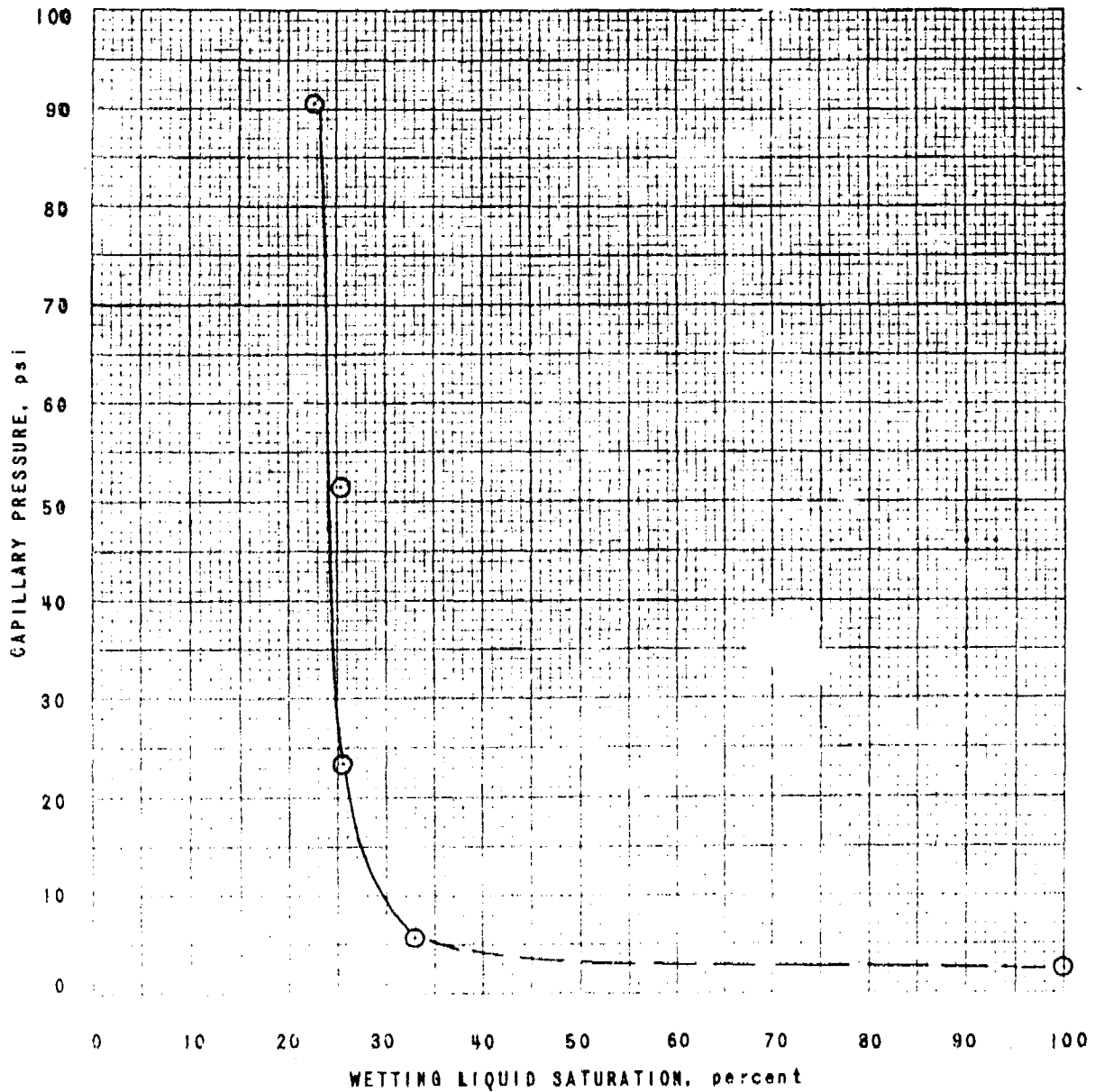


Figure 2
CAPILLARY PRESSURE CURVE

FIELD Angsi
RESERVOIR _____
WELL No. 1
DEPTH, ft 8037

CORE AN-36
PERMEABILITY, md 9.17
POROSITY, % 14.4
METHOD Centrifuge: Air-Kerosene

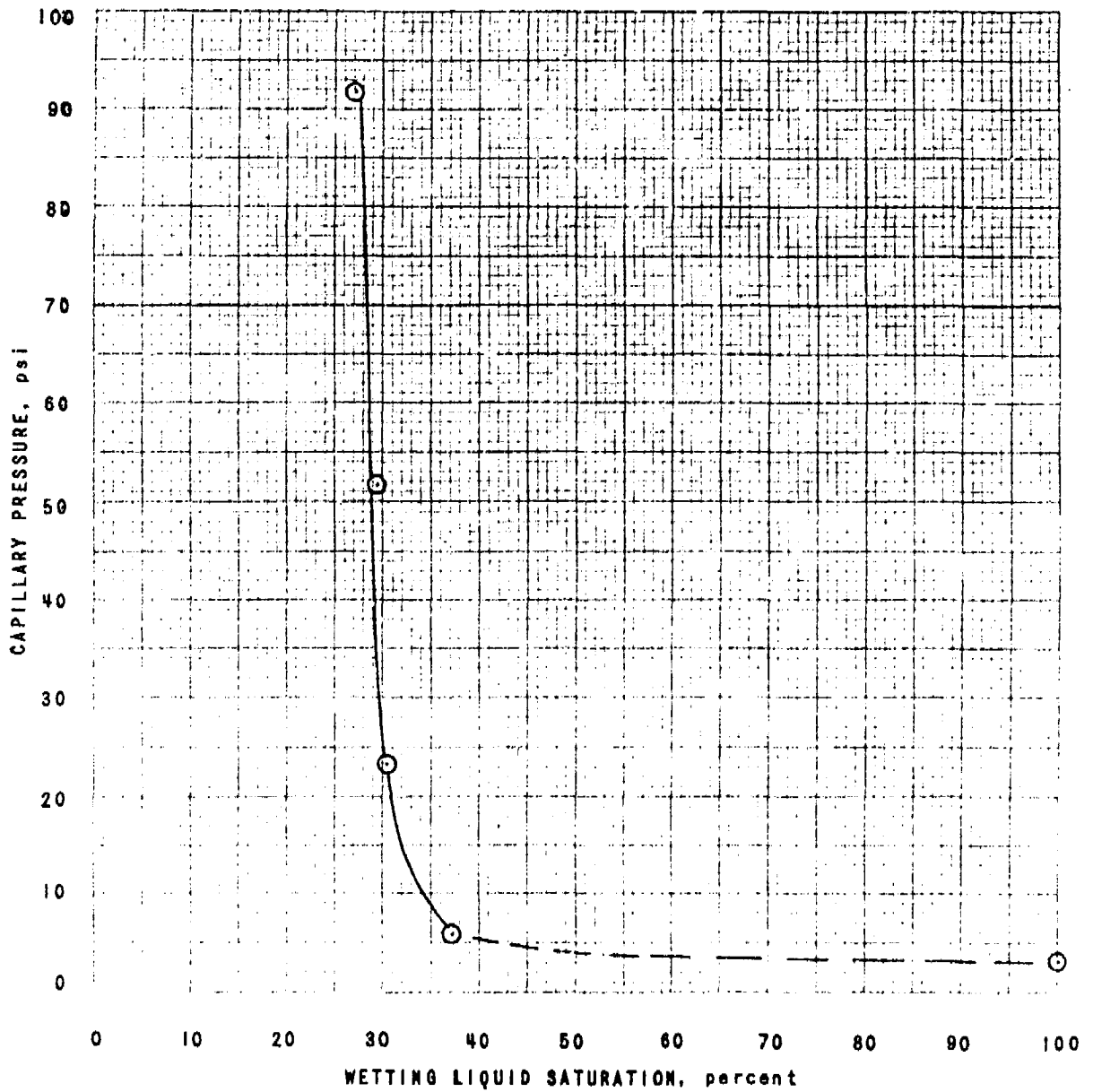


Figure 3
CAPILLARY PRESSURE CURVE

FIELD Angsi
RESERVOIR
WELL No. 1
DEPTH, ft 8038

CORE AN-37
PERMEABILITY, md 51.8
POROSITY, % 15.9
METHOD Centrifuge: Air-Kerosene

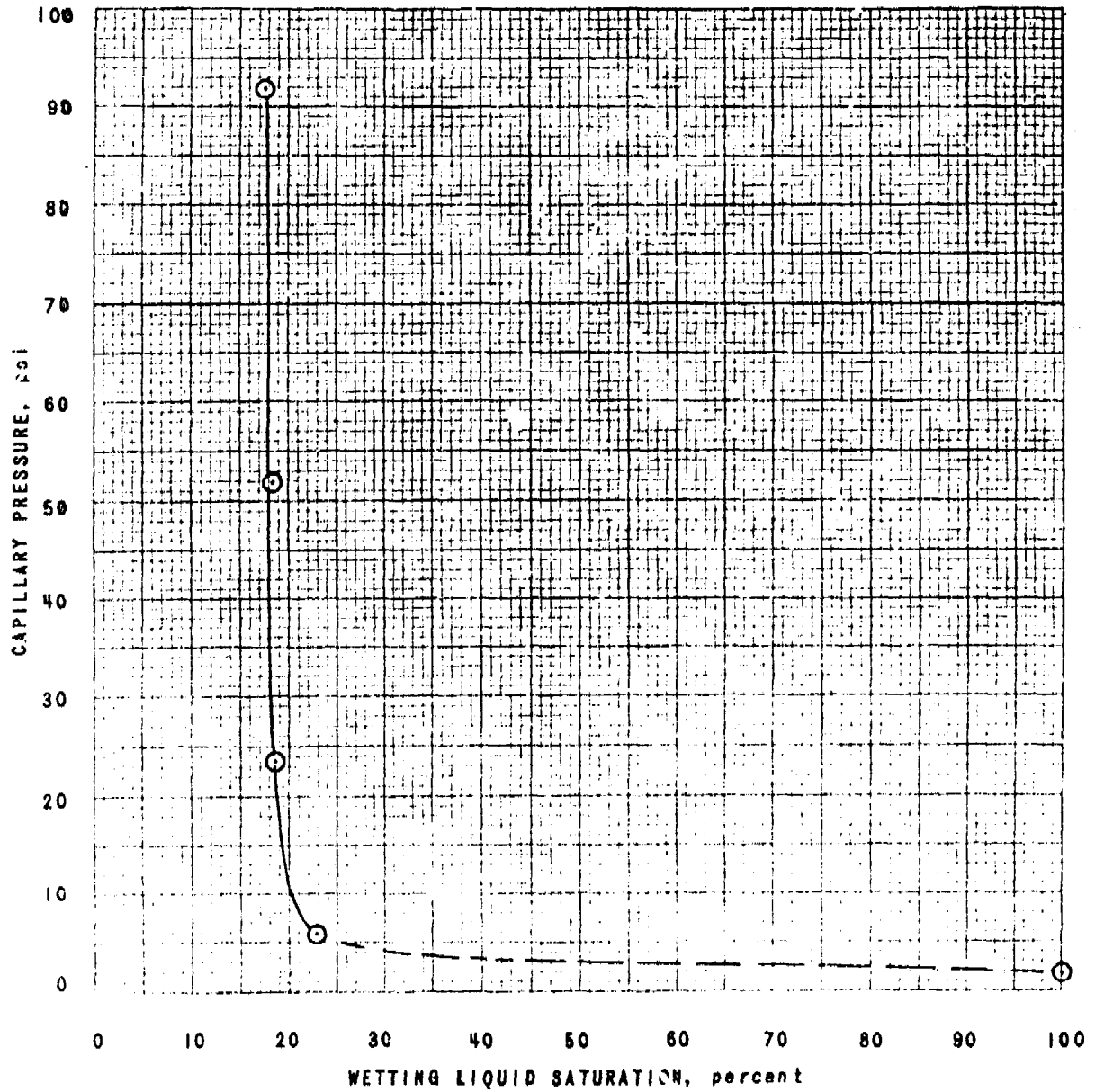


Figure 4
CAPILLARY PRESSURE CURVE

FIELD Angsi
RESERVOIR _____
WELL No. 1
DEPTH, ft 8040

CORE AN-39
PERMEABILITY, md 37.1
POROSITY, % 16.4
METHOD Centrifuge: Air-Kerosene

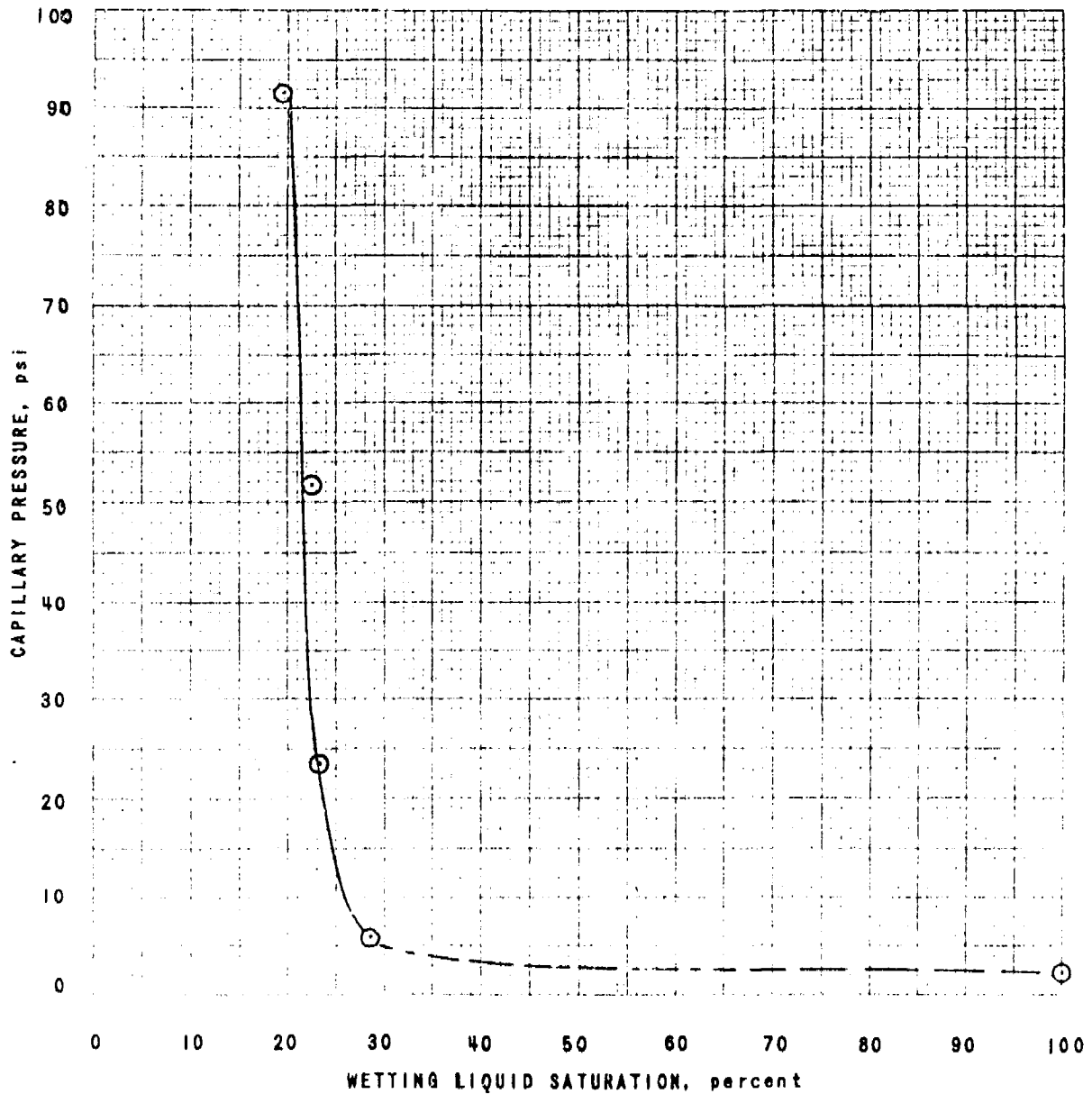


Figure 5

RESISTIVITY RATIO VS WATER SATURATION

FIELD	Malaysia	CORE	AN-28
RESERVOIR		DESATURATION EXPONENT	2.11
WELL	Angsi #1	FORMATION FACTOR	42.8
DEPTH, ft	8012	POROSITY	16.3
WATER TYPE	Lab Synthetic Brine	PERMEABILITY	4.42

NOTE: R/R_o = RATIO OF RESISTIVITY OF SAND SATURATED AT LEVEL UNDER CONSIDERATION TO RESISTIVITY OF SAND SATURATED 100%.

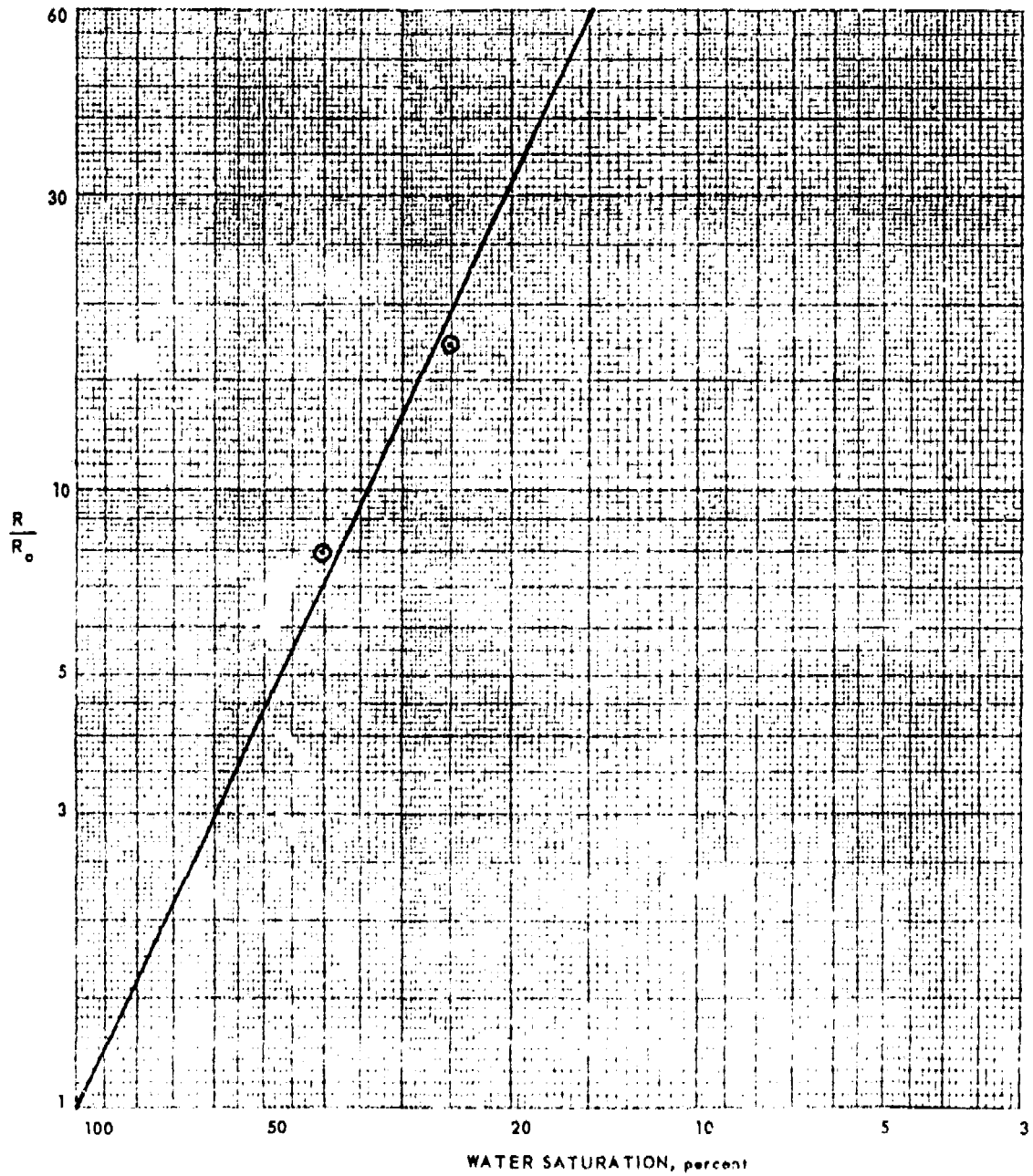


Figure 6

RESISTIVITY RATIO VS WATER SATURATION

FIELD <u>Malaysia</u>	CORE <u>AN-29</u>
RESERVOIR _____	DESATURATION EXPONENT <u>2.01</u>
WELL <u>Angsi #1</u>	FORMATION FACTOR <u>58.01</u>
DEPTH, ft <u>8013</u>	POROSITY <u>14.1</u>
WATER TYPE <u>lab Synthetic Brine</u>	PERMEABILITY <u>2.45</u>

NOTE: R/R_o = RATIO OF RESISTIVITY OF SAND SATURATED AT LEVEL UNDER CONSIDERATION TO RESISTIVITY OF SAND SATURATED 100%.

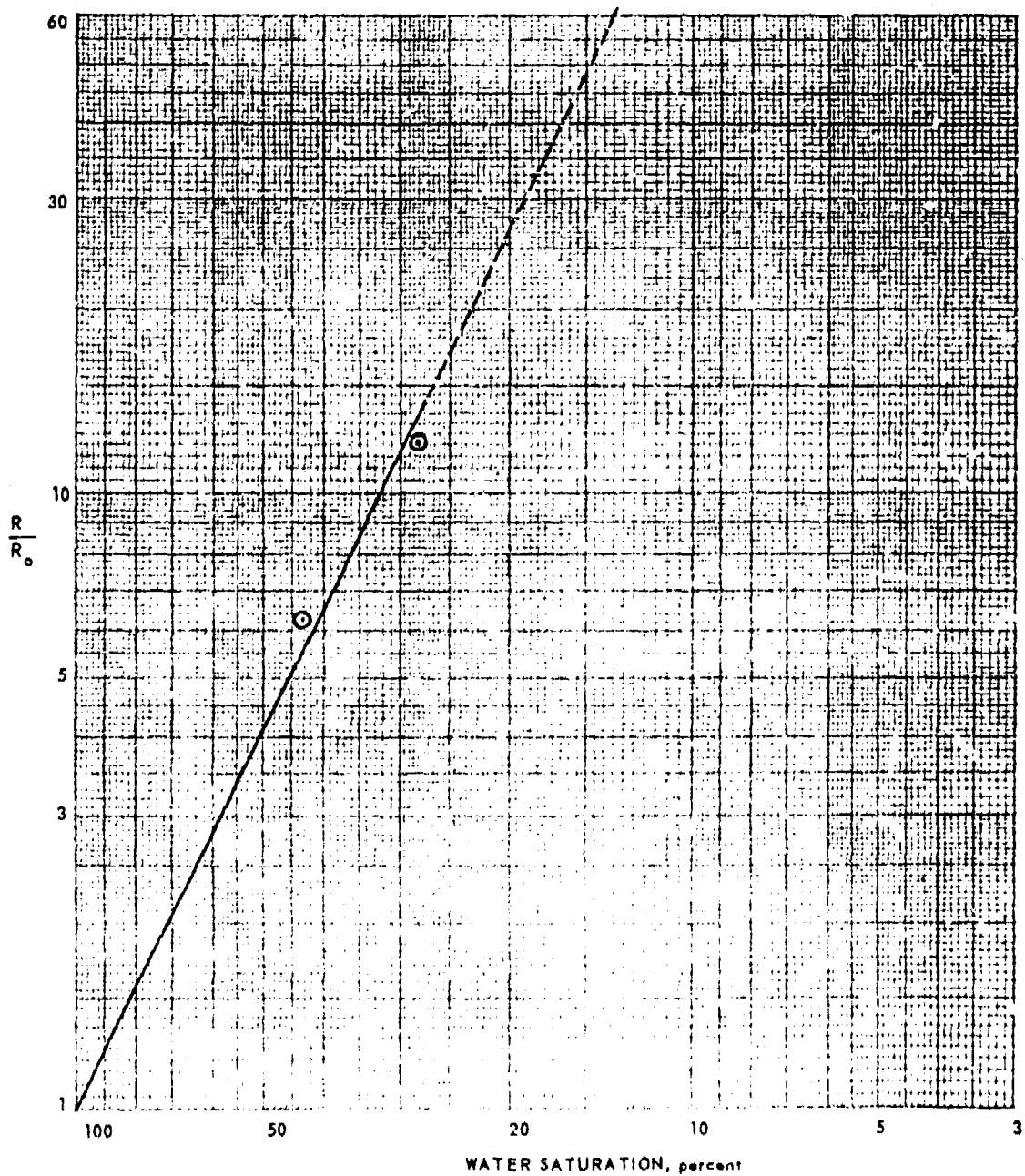


Figure 7

RESISTIVITY RATIO VS WATER SATURATION

FIELD <u>Malaysia</u>	CORE <u>AN-32</u>
RESERVOIR <u> </u>	DESATURATION EXPONENT <u>1.94</u>
WELL <u>Angsi #1</u>	FORMATION FACTOR <u>137.6</u>
DEPTH, ft <u>8029</u>	POROSITY <u>9.5</u>
WATER TYPE <u>Lab Synthetic Brine</u>	PERMEABILITY <u>0.10</u>

NOTE: R/R_0 = RATIO OF RESISTIVITY OF SAND SATURATED AT LEVEL UNDER CONSIDERATION TO RESISTIVITY OF SAND SATURATED 100%.

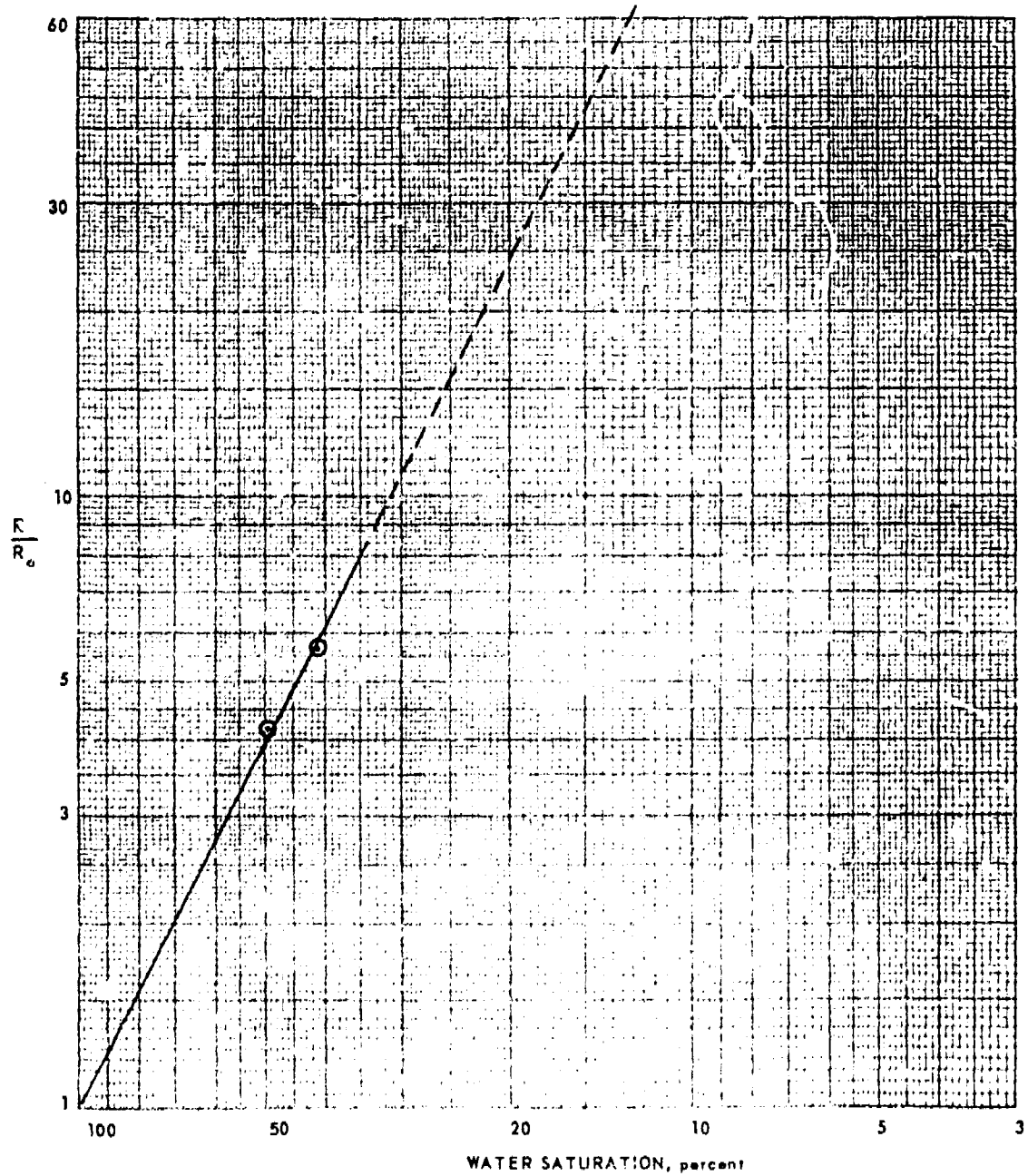


Figure 8

RESISTIVITY RATIO VS WATER SATURATION

FIELD	Malaysia	CORE	AN-32
RESERVOIR		DESATURATION EXPONENT	2.45
WELL	Angsi #1	FORMATION FACTOR	43.6
DEPTH, ft	8039	POROSITY	15.3
WATER TYPE	Lab Synthetic Brine	PERMEABILITY	37.1

NOTE: R/R_0 = RATIO OF RESISTIVITY OF SAND SATURATED AT LEVEL UNDER CONSIDERATION TO RESISTIVITY OF SAND SATURATED 100%.

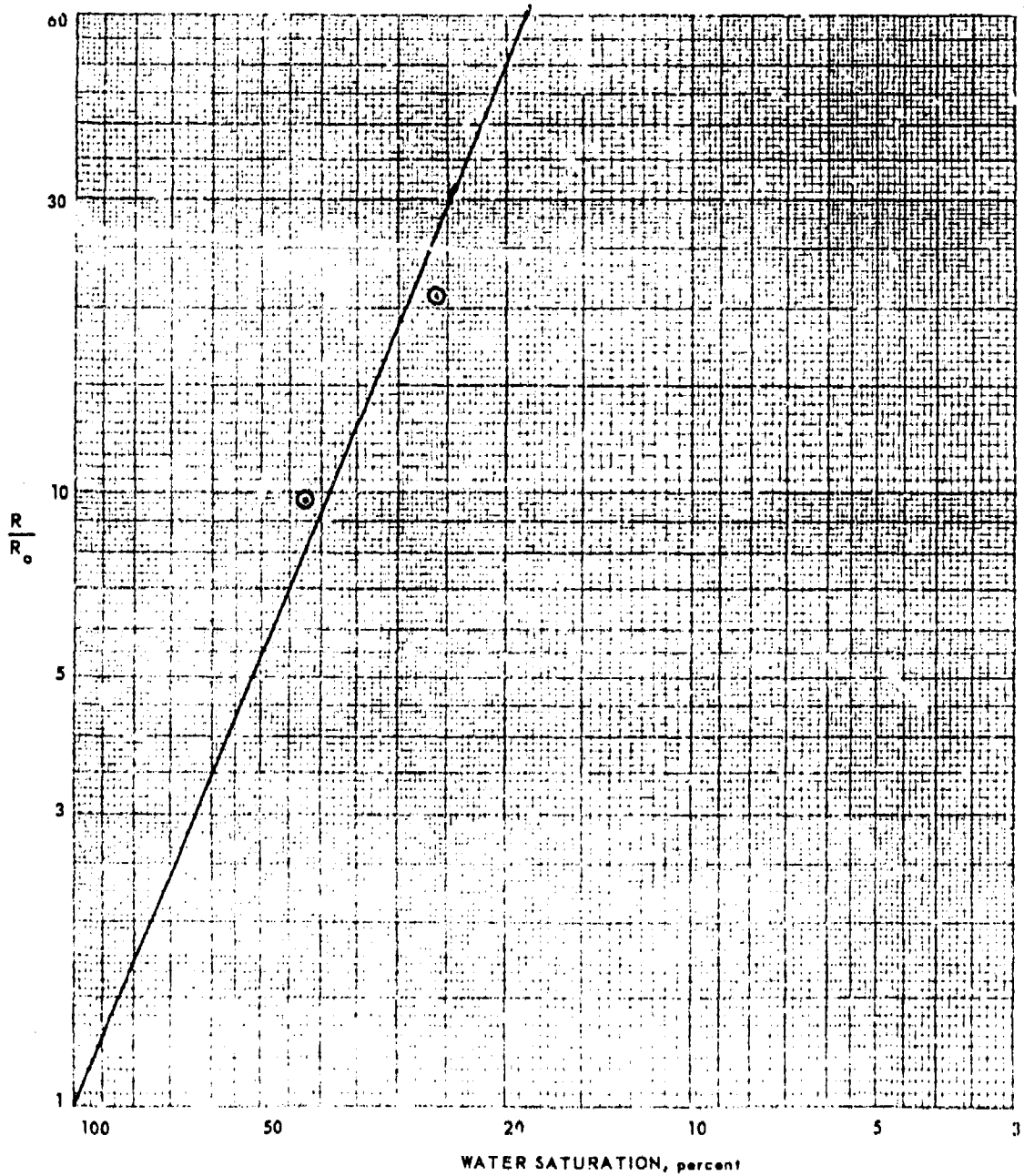


Figure 9

FORMATION FACTOR VS POROSITY

